

Mathematical comparison of Linguistic Categories without Universal Spines

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1 Neo-Emergentist Features

1.1 Against innate features

- Three factors model of Chomsky (2005).
 1. Universal Grammar (UG).
 2. Primary Linguistic Data (PLD).
 3. Principles not domain-specific to language.
- Radically impoverished UG.
- Not all features are innate.
 - As the number of languages studied increases, the required number of descriptive features continues to grow beyond evolutionary plausibility.
 - Sign language phonology - evolutionary problems given the relative rarity of hereditary deafness (Mielke, 2008).
 - Feature specifications pattern in contradictory ways across languages, (1).

(1) a.

ev ‘house’	evler ‘houses’
çocuk ‘child’	çocuklar ‘children’
fil ‘elephant’	filler ‘elephants’

Turkish vowel harmony, plural suffix *-ler/-lar*

b.

ember ‘person’	emberekkal ‘with the people’
fa ‘tree’	fákkal ‘with the trees’
híd ‘bridge’	hidakkal ‘with the bridges’

Hungarian vowel harmony, instrumental plural suffix *-ekkel/-akkal*

/i/, the most prototypical example of a front vowel (Backley, 2011), patterns as expected in Turkish, but as a back vowel in Hungarian (Törkenczy, 2011)¹.

- Non innateness of features suggested by i.a. Corina and Sagey (1989); Dresher (2003); Hall (2007); Mielke (2008); Biberauer (2011); Cowper and Hall (2014); Ramchand and Svenonius (2014); Wiltschko (2014).
- So not a recent idea, but these works are fairly independent, and innatist views still prevail.
 - (Chomsky, 2001, pg. 10) “We adopt **the conventional assumption** that L (a given language) makes a one time selection [F_L] from F (an innate set of features)” - emphasis mine.
 - (Biberauer, 2019, pg. 47-48) Biberauer 2019, pg. 47-48: “An informal survey of generative colleagues of all ages also suggests that **a great many remain committed to the necessary correctness of Chomsky’s (2001: 10) proposal**”.

¹This particular case can be explained diachronically, but there is a synchronic mismatch.

2 Features and DST

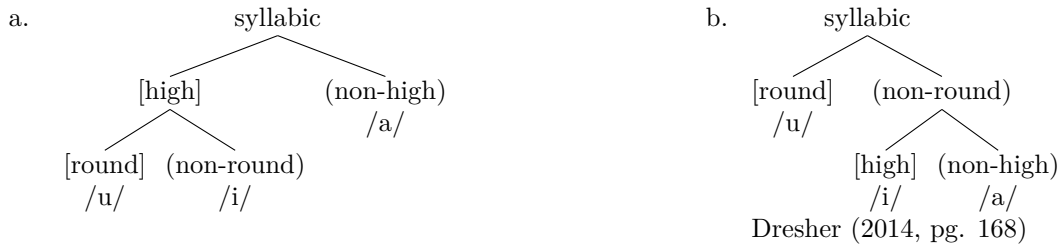
2.1 Dynamical Systems Theory

- Formally, a Dynamical System is a tuple $\langle T, X, \Phi \rangle$, where T is a set of times, X is a state space, and Φ is a transition function mapping changes in states across time.
 - $\Phi: X \times T \rightarrow X$.
 - The function takes a state and a time, and returns a new state.
- I follow the presentation of properties given in Bosch (2022) (which was presented at the ULAB Conference in 2022!).

2.2 Emergence

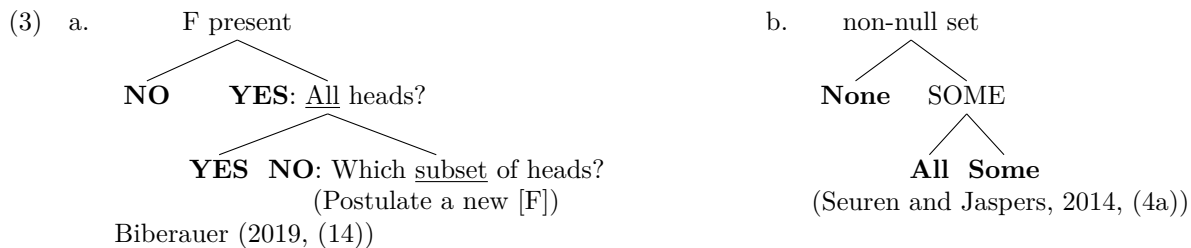
- The whole is bigger than the sum of the parts.
- This differs from ‘maturational’ accounts, eg. root infinitive hypothesis (Rizzi, 1993), which suggest time-based development of innate categories, but the whole is the sum of the parts Bosch (2022).
- Formalism mergence in linguistics - Successive Division (SD) (Dresher, 2003) and the None>All>Some learning path (Seuren and Jaspers, 2014; Biberauer and Roberts, 2015).

(2)



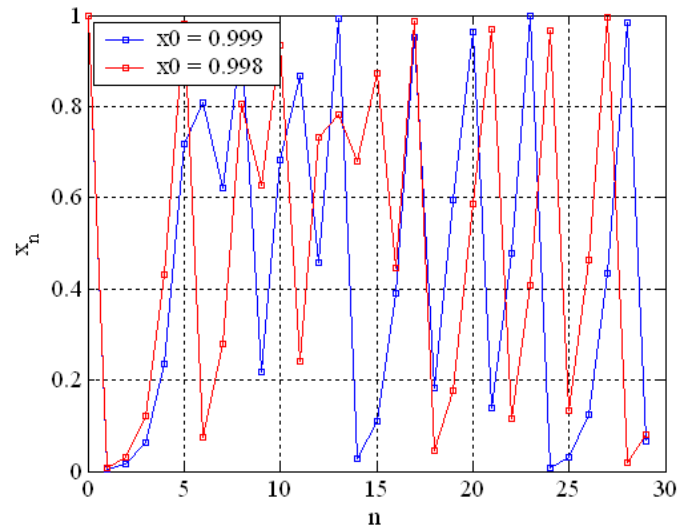
- (2) indicates two distinct possibilities for SD to generate a basic tripartite vowel division.
- These different divisions should have different emergent properties, relating to phonological processes like sound change, vowel reduction etc.
- There is support for this approach on typological grounds (Bazalgette, 2015).

Biberauer’s formalisation (3a) is inspired by Seuren and Jaspers (2014)’s work on concept formation (3b).



- This process of division is found in other cognitive domains, as investigated by Biberauer and Bosch (2021).
- One example is vision Kidd et al. (2012), where formation of visual categories proceeds by identifying ‘edges’ of larger units and making use of contrast.
- As this occurs in non-domain specific cognition, it is plausibly a reflex of a Third Factor.

2.3 Sensitivity to Initial Conditions



The image above, from Madhekar (2011), shows a dynamical system over time. It maps the equation $x_{n+1} = 4x_n(1-x_n)$, for two values of x_0 : 0.999 (blue) and 0.998 (red). A tiny change in the initial conditions leads to massive, non-linear divergences in later states.

The same concept can be seen in feature systems:

- Early vs late acquired properties (Wexler, 1998; Tsimpli, 2014)
- Modality differences.

So what are the initial conditions for language (or what is innate)?

- Phonology: are vowels and consonants a candidate for innateness?
 - Nespor et al. 2003 - vowels and consonants have a division of labour. Consonants mark lexical units, vowels are used in grammatical properties.
 - * eg. **y cn prbbl rd lxcl nts wtht vwls, u iou oaa i u ae.**
(you can probably read lexical units without vowels, but without consonants it's much harder).
 - * The role of vowels in inflection - sing / sang / sung.
 - Vowels and consonants are also argued for sign languages - articulator movements pattern like vowels and handshapes/locations like consonants (Liddell 1984).
 - So vowels and consonants are emergent, and not based on innate articulatory/acoustic/auditory properties!
- Radically impoverished UG in syntax - Wiltschko (2014)'s Universal Spine (US)?:

(4) a. Classification $\leq$ Point-of-view $\leq$ Anchoring $\leq$ Linking

b. $V \leq v \leq T \leq C$ (Conventional equivalents in verbal domain in English)

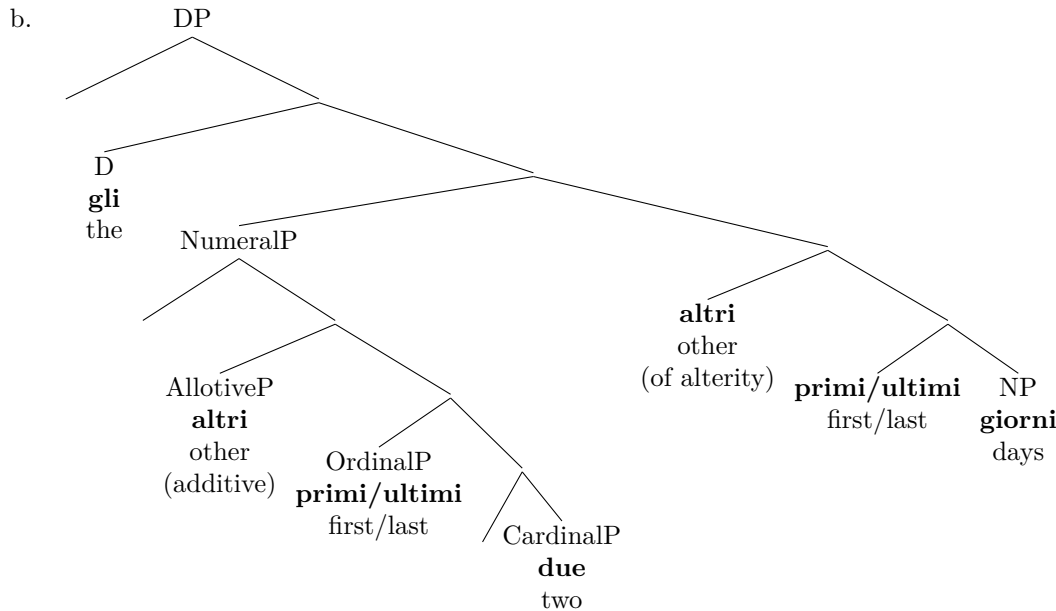
- This is proposed as an innate semiotic hierarchy, to which language specific linguistic categories can be attached.
- I argue that at this point, we are better off taking the stronger hypothesis, no substantive UG.
- By SD, the segments of the US could emerge as salient information in the world.
- However, a US presupposes that **all** language users have these divisions as basic.
- Challenges for this:
 - * Modality
 - * Parts-of-speech
 - * Uniformity
- It is worth investigating the stronger hypothesis, **an initial state of no (substantive) UG**.

2.4 Self-Similarity

- Self-similarity refers to repetition of similar processes at similar scales, leading to seemingly infinite complexity.
- eg. snowflakes in the natural world.
- In syntax, self-similarity can be seen with Extended Projection (EP)s (Grimshaw, 1991), which represent a lexical head with its related functional heads.

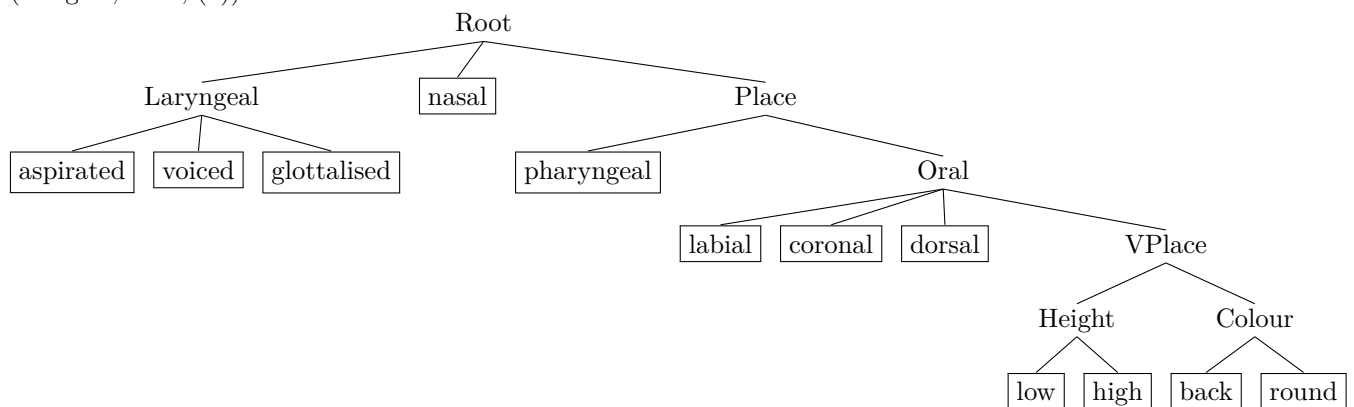
(5) (Cinque, 2022, (29))

a. Gli altri primi/ultimi due altri primi/ultimi giorni.
‘The other first/last two other first/last days.’



- Acquisitional support (Bosch, 2023).
- Phonological equivalent - feature geometries:

(6) (Padgett, 2002, (3))



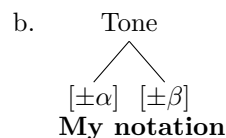
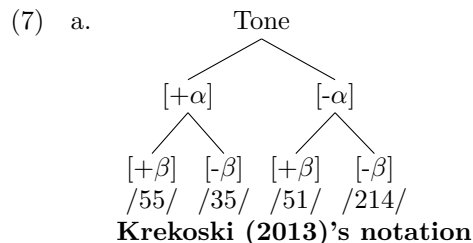
- Granularity - larger objects can be viewed as both minimal units and as having internal structure - Category Theory!

2.5 Edge-of-chaos

- Linguistic systems allow emergence and variety.
- But are learnable avoid ‘rampant and unconstrained variation’ (Biberauer, 2019, pg. 59).
- Edge-of-chaos - well attested in DST and human cognition (eg. Waldrop, 1993).

3 Comparison of features with Successive Division

- Analysis of Standard Mandarin tones.
- $[-\alpha]$ is a falling tone, $[-\beta]$ is a rising tone.
- The two features are independent of one another, in terms of permutations and temporal relations.



More complicated examples - Vowel Inventories:

(8) a.

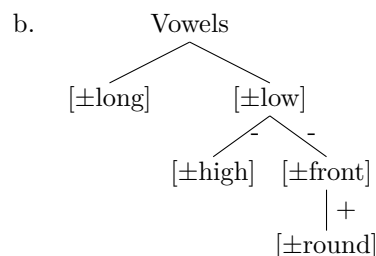
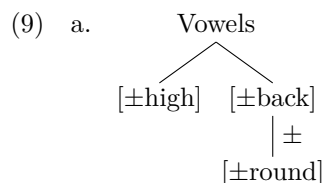
	Front		Back	
	Unround	Round	Unround	Round
High	i	ü	ɪ	u
low	e	ö	a	o

Turkish vowel inventory

b.

	Front		Back
	Unround	Round	
High	i í	ü ú	u ú
Mid	e é	ö ő	o ó
Low			a á

Hungarian vowel inventory



- Avoids massive overgeneration.
- Limited permutations - contrastive backness for front vowels only in Hungarian.
- Temporal relations:
 - Perception (Stevens et al., 1986)
 - Phonological behaviour: Turkish has one backness harmony alternation between /a/ and /e/ (1a), and another ‘back + round’ness alternation between /i/, /ü/, /ɪ/ and /u/.
 - Acquisition:

(10)

Vowel	a	e	i	o	ö	u	ü
%	93	90	96	91	66	81	69

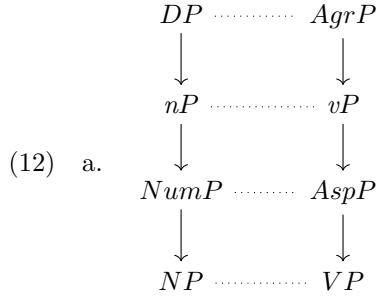
Zajdó and Stoel-Gammon (2003), accuracy of Hungarian girls age 2;0 in producing short vowels.

4 Order Preservation

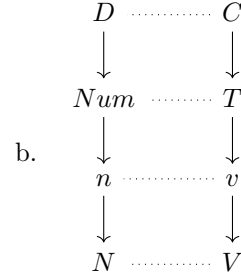
Comparison of verbal domain and nominal one:

- (11) Innovation in Belfast English for some speakers, (Henry, 2005, 1610-1611, cited in Harbour 2020):
 a. ‘The (two) men is talking.’

- Innatist approaches: - ‘**presumed parallelisms** between the noun phrase and the clause’ (Salzmann, 2020, pg. 1)



(Megerdooian, 2008, (42))



(Roberts, 2024, (21))

The verbal domain and nominal domain can be of different sizes:

- (13) Greenlandic (Eskimo-Aleut) (Fortescue, 1984, pg. 137):

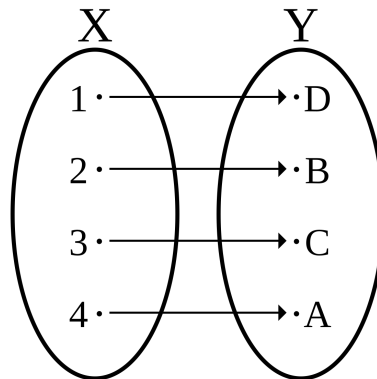
Kaalat tikik-nngik-kallar-puq

K. arrive-yet/still-NEG-IND:3S

‘Kaalat still has not arrived.’

- So comparison between the verbal and nominal domains cannot universally be a bijection/isomorphism.
- These terms generalise one-to-one functions.
- In the image below, for functions $f: X \rightarrow Y$ and $g: Y \rightarrow X$, a ‘round-trip’ always leads to the original value, eg. $gf(x) = x$, $fg(y) = y$ for $x \in X$, $y \in Y$

from Wikipedia: bijection



- Weaker form of comparison - Adjunctions².
- Adjunctions are “perhaps the most profound contribution” of Category theory to “the history of general mathematical ideas” (pg. 438 Goldblatt, 2006).

²I follow Song (2019) in capitalising this term to avoid confusion with the unrelated linguistic concept.

- For categories and functors $F: \mathcal{A} \rightarrow \mathcal{B}$ and $G: \mathcal{B} \rightarrow \mathcal{A}$, F is left-Adjoint to G and G is right-Adjoint to F if there is a hom-set isomorphism $\mathcal{B}(F(A), B) \cong \mathcal{A}(A, G(B))$ which is natural for $A \in \mathcal{A}$ and $B \in \mathcal{B}$:

$$(14) \quad \begin{array}{ccc} & F & \\ \mathcal{A} & \xrightarrow{\quad} & \mathcal{B} \\ & \perp & \\ & G & \end{array}$$

- This looks very complicated! It's equivalent in partially ordered set categories is a Galois connection, which also looks complicated, but is more intuitive:
- For Partially Ordered Sets A and B , a Galois connection between them consists of two monotonic functions $f: A \rightarrow B$ and $g: B \rightarrow A$ such that for all $a \in A$ and $b \in B$:

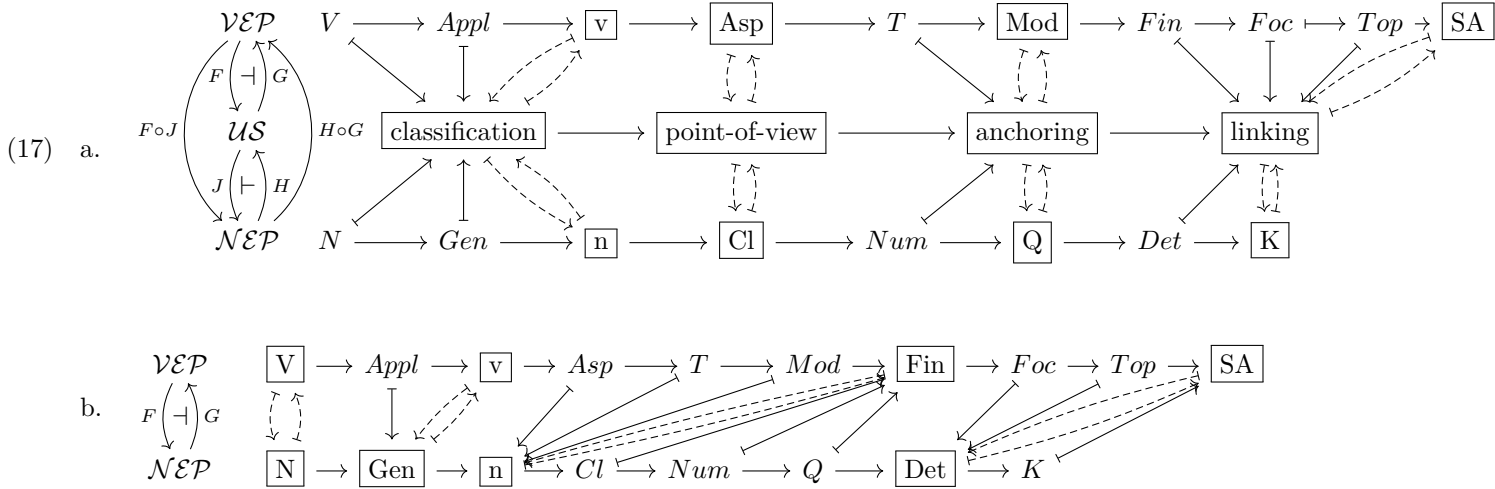
$$(15) \quad f(a) \leq b \text{ iff } a \leq g(b).$$

- If we look at round trips again, elements in A can only get bigger or stay the same, whereas elements in B can only get smaller or stay the same.

$$(16) \quad \text{a. } a \leq g(f(a))$$

$$\text{b. } f(g(b)) \leq b$$

- Song's example of Adjoint comparison with a US (17a), vs mine without (17b).
- Role of asymmetry in determining comparisons - avoiding chaos?



- Uses outside of core categorial projections - monotonicity phenomena (Biberauer, 2017).
- Adjunctions can model interactions between two different objects which have order relations, when these objects are of different sizes.
- Objects are not purely linguistic - linear order, height of articulation (Bross, 2020) - Third Factor? Eg. order presentation between presence of infinitival *-e* and verb order in Italian with object clitics (Cardinaletti and Shlonsky, 2004, cited in Biberauer (2018)):

- (18) a. *Li ho voluti fare andare a prendere a Maria.*
 them I.have want.PP make.INF go.INF to deliver.INF to Maria
 ‘I wanted to make Maria go and fetch them.’
 b. *Li ho voluti far andare a prendere a Maria.*
 c. *Li ho voluti far andar a prendere a Maria.*
 d. **Li ho voluti fare andar a prendere a Maria.*

Acronyms

Agr Agreement. 6

Appl Applicative. 7

Asp Aspect. 6, 7

Cl Classifier. 7

Det Determiner. 7

DST Dynamical Systems Theory. 2, 5

EP Extended Projection. 4

Fin Finite. 7

Foc Focus. 7

Gen Genitive. 7

K Case. 7

Mod Modality. 7

NEP Nominal Extended Projection. 7

Num Number. 6, 7

PLD Primary Linguistic Data. 1

poset Partially Ordered Set. 7

Q Quantifier. 7

SA Speech Act. 7

SD Successive Division. 2, 3, 5

Top Topic. 7

UG Universal Grammar. 1, 3

US Universal Spine. 3, 7

VEP Verbal Extended Projection. 7

VPlace Vowel Place. 4

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